Pavithra PM Nair

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Research Interests

Responsible AI • Fairness & Bias in LLMs • Cross-Lingual & Cultural Robustness • AI for High-Stakes Domains (Legal, Finance) • Human–AI Decision-Making • AI Governance

Education

Amrita Vishwa Vidyapeetham — Kerala, India
Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence) — *2020 - 2024*
CGPA: 9.36/10 • ***Top 4% of cohort*** *(Rank 6/154)*

Ben-Gurion University of the Negev — Beer-Sheva, Israel
Machine Learning and Cybersecurity Summer School • ***Full Scholarship*** — *August - September 2023*

Experience

Amrita Vishwa Vidyapeetham
Scientist — *April 2026 - Present*

- **LLM Evaluation** – Lead a team of 3 researchers developing TrackLLM (track-llm.in), an evaluation framework for auditing LLM applications for bias, hallucinations, toxicity, and multilingual robustness; developed with the Department of Telecommunications (Govt. of India) and the IndiaAI Mission.
- **Responsible AI Research** – Conduct research on fairness in LLMs and AI safety, including demographic bias in economic advice, financial hallucinations in scam contexts, and bias drift after fine-tuning.

Tata Research Development and Design Centre (TCS Research)
Researcher — *August 2024 - April 2026*

- **Regulatory Compliance Automation** - Built Reg2Req, an automated LLM pipeline that identifies requirement-bearing clauses in regulations and derives traceable, plain-language software requirements, evaluated on the full GDPR and EU AI Act (RE 2026 Industrial Innovation Track); co-developed C-PASS, an organization-centric framework for compliance and penalty assessment (Information and Software Technology).
- **Ambiguity Handling** - Developed a novel approach for detecting and resolving pragmatic ambiguities in natural language requirements using multi-level domain knowledge bases; published at EASE 2026.

Prodago - AI Governance (as contracted through Traboda Cyberlabs)
Researcher — *November 2023 - August 2024*

- **AI Governance Assistant** - Co-designed a RAG assistant for navigating AI governance regulations with domain experts, and set up Ragas-based evaluation workflows to benchmark and select models.

Association for Computing Machinery
Research Intern — *July - August 2023*

- **Bias Detection & Mitigation** - Analyzed demographic biases in computer vision datasets and contributed to a mitigation approach combining data augmentation and reweighing.

Publications

† denotes a researcher mentored by me

Conference & Journal

1. From Regulation to Requirements: An Automated Requirement Derivation and Explanation Pipeline
   IEEE International Requirements Engineering Conference (RE), Industrial Innovation Track 2026
   Pavithra PM Nair, Preethu Rose Anish

2. Detecting and Resolving Pragmatic Ambiguities in Natural Language Requirements Using Multi-Level Domain Knowledge Bases
   Evaluation and Assessment in Software Engineering (EASE) 2026
   Pavithra PM Nair, Preethu Rose Anish

3. Sruthi: What Becomes Sayable Through Peer-Mode Reflective Scaffolding in Fieldwork Communication
   ACM SIGCHI Conversational User Interfaces (CUI) 2026 ***(Best Paper Honourable Mention)***
   Srividya Sheshadri, Manju Balakrishna Pillai, Renuka Kumar, **Pavithra PM Nair**, Apoorv Gupta, Megha Tm, Sooraj S Nair, Balu M Menon, Unnikrishnan Radhakrishnan, Bhavani Rao R

4. C-PASS: An Organization-centric Framework for Compliance and Penalty Assessment Using Large Language Models
   Information and Software Technology 2026
   Gokul Rejithkumar, Sachin Pawar, **Pavithra PM Nair**, Nitin Ramrakhiyani, Preethu Rose Anish

5. To Solicit or Not to Solicit? Impact of AI Assistance Delivery Mechanisms on Decision-Making
   International Journal of Human–Computer Interaction 2025
   Pavithra PM Nair, Gilad Gressel, Malavika Nambrath Anand, Krishnashree Achuthan

6. An Integration of Indian Philosophy and Machine Learning for Human-like Cognition
   International Conference on Automation, Computing and Renewable Systems (ICACRS) 2023
   Akhbar Sha, Anvita Reddy Inture, Jose Joseph, **Pavithra PM Nair**, MJ Sheeba, T Anjali

Workshop

7. "Sorry, Can't Help You": How Large Language Models Judge Failures to Help Across Languages
   C3NLP Workshop @ ACL 2026 ***(Best Paper Award)***
   Pavithra PM Nair, Gilad Gressel, Krishnashree Achuthan

8. Lost in Translation? How Language Shapes Responsibility Attribution in Large Language Models
   C3NLP Workshop @ ACL 2026
   Pavithra PM Nair, Gilad Gressel, Krishnashree Achuthan

9. Less Abstention, More Bias? Demographic Bias Drift After Supervised Fine-Tuning
   NeurIPS 2026 Workshops (Pre-Training to Post-Training; Foundations of LLM Post-Training in Changing Environments)
   Seema Guruvadoo[†], **Pavithra PM Nair**, Gilad Gressel, Krishnashree Achuthan

10. Vichara: Appellate Judgment Prediction and Explanation for the Indian Judicial System
    AI and Law Workshop @ AAAI 2026
    Pavithra PM Nair, Preethu Rose Anish

Under Review

11. RupeeBias: Auditing Demographic Bias in Indian Economic Guidance from Large Language Models
    ACL Rolling Review 2026
    Pavithra PM Nair, Bhavik Talaviya[†], Shourya Bhushan[†], Rahul Pankajakshan, Seema Guruvadoo[†], Avinash Agarwal, Gilad Gressel, Krishnashree Achuthan

12. FauxFinance: When LLMs Explain Financial Terms That Do Not Exist
    ACL Rolling Review 2026
    Bhavik Talaviya[†], **Pavithra PM Nair**, Gilad Gressel, Krishnashree Achuthan

Awards and Honors

- **Best Paper Award (2026)** – C3NLP Workshop @ ACL 2026
- **Best Paper Honourable Mention (2026)** – ACM SIGCHI Conversational User Interfaces (CUI)
- **National Launch Selection (2026)** – TrackLLM selected for national launch by the CEO of the IndiaAI Mission ($1.2B national AI program) and UIDAI (operator of Aadhaar, 1.3B+ users)
- **Cited in the Parliament of India (2026)** – TrackLLM cited by the Minister of State for Electronics & IT in a written reply to the Lok Sabha (Tech Observer)
- **Economic Times Campus Star (2023)** – Selected through a four-round national evaluation of technical and professional skills
- **Full Scholarship, Ben-Gurion University of the Negev (2023)** – Summer School in Machine Learning & Cybersecurity
- **Academic Achievement Award (2023)** – Amrita Vishwa Vidyapeetham, for outstanding academic performance in the B.Tech. CSE (AI) program
- **Finalist, European Rover Competition (2021)** – Developed remote-navigation software for a rover in a simulated Mars terrain

Extracurriculars, Leadership and Initiatives

Google Developer Student Club
Lead *2023 - 2024*

- **Technical Knowledge Sharing** – Conducted hands-on workshops on topics such as web development, AI, and cloud computing, emphasizing practical skills and real-world applications.
- **Community Building** - Grew the chapter to 40+ members and ran a peer mentorship program pairing beginners with experienced members for hands-on projects.

amFOSS – Computer Science Club (amfoss.in)
Mentor & Member *2020 – 2024*

- **Open Source Contribution** – Contributed to open source projects both within the club (amFOSS CMS) and external projects (Navidrome, Ersilia Open Source Initiative).
- **Podcast** – Led ideation for and co-hosted the amFOSS Podcast to promote open-source engagement.